

52. 2D Drawing Engine (DRW2D)

52.1 Overview

This MCU incorporates a 2D drawing engine (DRW2D). Table 52.1 lists the DRW2D specifications.

Table 52.1 DRW2D Specifications

Item		Descriptions
Vector drawing engine		Extended rendering primitives supported in hardware • Lines • Polygons • Circles and ellipses • Quadratic Béziers • Texture mapping
BitBLT function		Seven types of BitBLT operations • Fill • Copy • Stretch BitBLT • Rotate and scale • Alpha blending • Bilinear filtering • Color conversion
Color format	Frame buffer	Four types • A(8) • RGB(565) • ARGB(4444) • ARGB(8888)
	Texture	11 types • CLUT(1) • CLUT(2) • CLUT(4) • CLUT(8) • A(8) • ACLUT(44) • RGB(565) • ARGB(4444) • ARGB(1555) • RBG(888) • ARGB(8888)
Texture		• Run-length encoding (RLE) unit • 256-entry color look-up table (CLUT) • Chroma keying unit
Render		Two modes for rendering process • Register mode • Display list mode
Data arrangement		Little endian
Performance counter		• Two independent 32-bit counters • Trigger is selectable from 14 events
Interrupt sources		Two interrupt sources • Enumeration finished interrupt (DRWENUIR) • Display list finished interrupt (DRWDLIR)

The DRW2D provides two inputs (texture read and frame buffer read) and one output (frame buffer write). The internal color format is always ARGB(8888). The color formats from the inputs are converted to the internal format on read and converted back on write.

Figure 52.1 shows examples of objects that can be drawn in hardware with the DRW2D and Figure 52.2 shows the block diagram of the module.

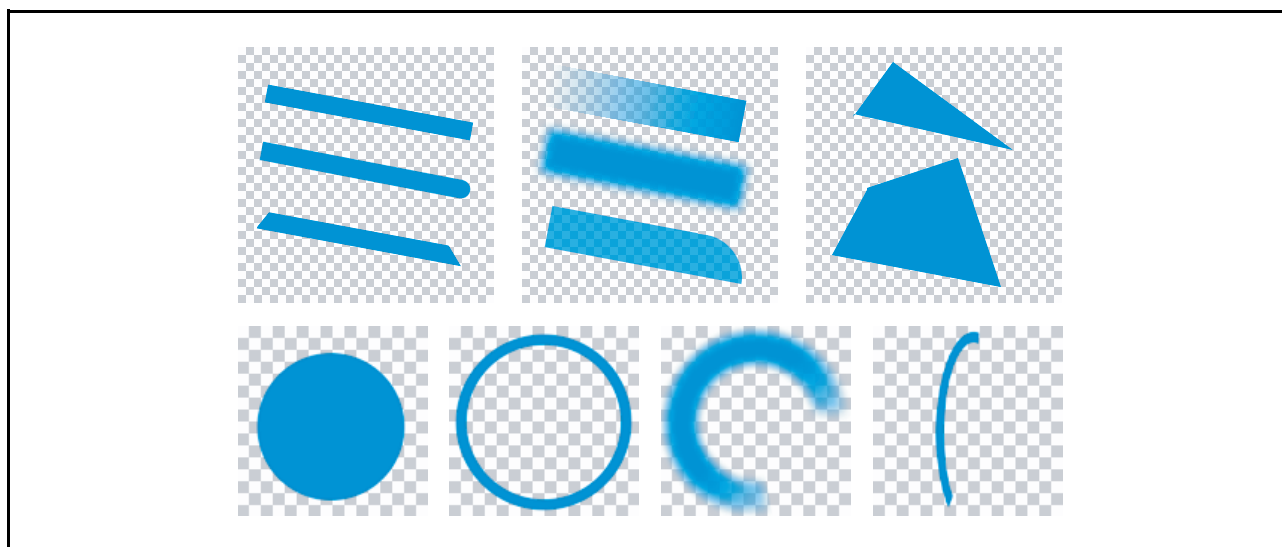


Figure 52.1 Examples of Drawing Objects

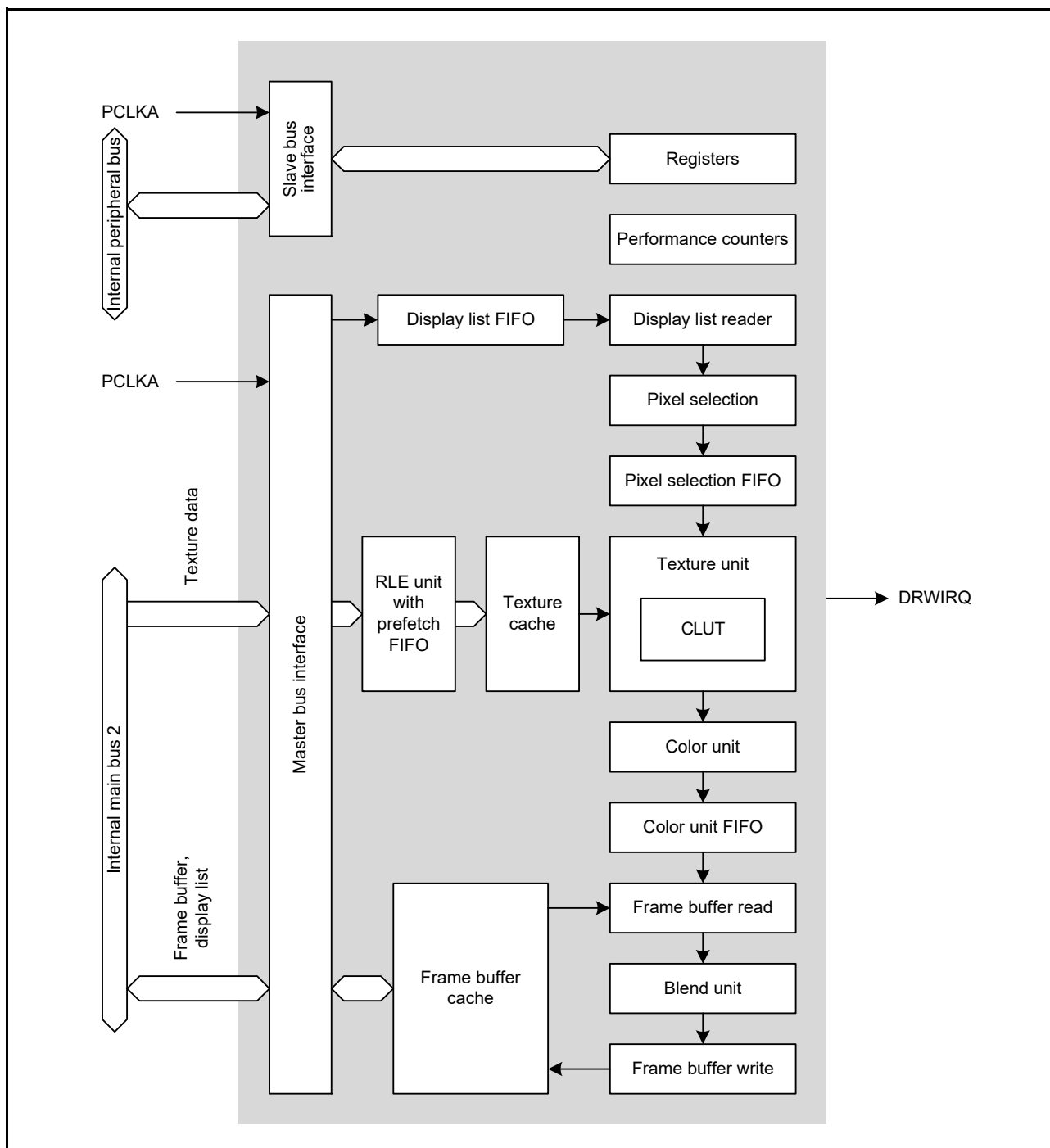


Figure 52.2 Block Diagram of the DRW2D

The DRW2D accesses internal main bus 2 as bus master through each caches for:

- Reading and writing pixel data from and to the frame buffer
- Reading textures
- Reading display lists

The registers are accessed through the internal peripheral bus.

52.2 Register Descriptions

52.2.1 Geometry Control Register (CONTROL)

Address(es): DRW2D.CONTROL 000E 3000h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	SPANS TR	SPAN BT	UNION CD	UNION AB	UNION 56	UNION 34	UNION 12	BAND2 EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	BAND1 EN	LIM6TH	LIM5TH	LIM4TH	LIM3TH	LIM2TH	LIM1TH	QUAD3 EN	QUAD2 EN	QUAD1 EN	LIM6E N	LIM5E N	LIM4E N	LIM3E N	LIM2E N	LIM1E N
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	LIM1EN	Limiter 1 Enable	0: Disable 1: Enable	W
b1	LIM2EN	Limiter 2 Enable	0: Disable 1: Enable	W
b2	LIM3EN	Limiter 3 Enable	0: Disable 1: Enable	W
b3	LIM4EN	Limiter 4 Enable	0: Disable 1: Enable	W
b4	LIM5EN	Limiter 5 Enable	0: Disable 1: Enable	W
b5	LIM6EN	Limiter 6 Enable	0: Disable 1: Enable	W
b6	QUAD1EN	Limiters 1 and 2 Quadratic Coupling Enable	0: Disable (Linear equation mode) 1: Enable (Quadratic equation mode)	W
b7	QUAD2EN	Limiters 3 and 4 Quadratic Coupling Enable	0: Disable (Linear equation mode) 1: Enable (Quadratic equation mode)	W
b8	QUAD3EN	Limiters 5 and 6 Quadratic Coupling Enable	0: Disable (Linear equation mode) 1: Enable (Quadratic equation mode)	W
b9	LIM1TH	Limiter 1 Threshold Mode Enable	0: Disable 1: Enable	W
b10	LIM2TH	Limiter 2 Threshold Mode Enable	0: Disable 1: Enable	W
b11	LIM3TH	Limiter 3 Threshold Mode Enable	0: Disable 1: Enable	W
b12	LIM4TH	Limiter 4 Threshold Mode Enable	0: Disable 1: Enable	W
b13	LIM5TH	Limiter 5 Threshold Mode Enable	0: Disable 1: Enable	W
b14	LIM6TH	Limiter 6 Threshold Mode Enable	0: Disable 1: Enable	W
b15	BAND1EN	Limiter 1 Band Filter Enable	0: Disable (half plane intersection) 1: Enable	W
b16	BAND2EN	Limiter 2 Band Filter Enable	0: Disable (half plane intersection) 1: Enable	W
b17	UNION12	Limiters 1 and 2 Combiner Operation Select	0: Combine limiters 1 and 2 as intersect (Output is called A) 1: Combine limiters 1 and 2 as union (Output is called A)	W

Bit	Symbol	Bit Name	Description	R/W
b18	UNION34	Limiters 3 and 4 Combiner Operation Select	0: Combine limiters 3 and 4 as intersect (Output is called B) 1: Combine limiters 3 and 4 as union (Output is called B)	W
b19	UNION56	Limiters 5 and 6 Combiner Operation Select	0: Combine limiters 5 and 6 as intersect (Output is called D) 1: Combine limiters 5 and 6 as union (Output is called D)	W
b20	UNIONAB	Outputs A and B Combiner Operation Select	0: Combine outputs A and B as intersect (Output is called C) 1: Combine outputs A and B as union (Output is called C)	W
b21	UNIONCD	Outputs C and D Combiner Operation Select	0: Combine outputs C and D as intersect (Output is final) 1: Combine outputs C and D as union (Output is final)	W
b22	SPANABT	Spanabort Enable	0: Disable 1: Enable	W
b23	SPANSTR	Spanstore Enable	0: Disable 1: Enable	W
b31 to b24	—	Reserved	The write value should be 0.	W

Note: This register is write-only and reading returns the value of the STATUS register. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.2 Surface Control Register (CONTROL2)

Address(es): DRW2D.CONTROL2 000E 3004h

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
RLEPIXW[1:0]	BDIA	BSIA	CLUTFORM	COLKEYEN	CLUTEN	RLEEN	WRALPHA[1:0]	WRFMT[1:0]	RDFMT[1:0]	TEXFILTY	TEXFILTX				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
TEXCLPY	TEXCLPX	BC2	BDI	BSI	BDF	BSF	WRFMT2	BDFA	BSFA	RDFMT2[1:0]	USEACB	PTNSRCL5	TEXENA	PTNEN	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PTNEN	Pattern Color Enable	0: Disable pattern 1: Enable pattern (Pixel source is a pattern color (blend of registers COLOR1 and COLOR2 depending on the PATTERN register and pattern index))	W
b1	TEXENA	Texture Enable	0: Disable texture 1: Enable texture (Pixel source is read from texture and used as an alpha to blend between registers COLOR1 and COLOR2)	W
b2	PTNSRCL5	Limiter 5 Pattern Index Enable	0: U-limiter is used as pattern index 1: Limiter 5 is used as pattern index	W
b3	USEACB	Alpha Blend Mode Select	0: Use write-alpha mode (The frame buffer format should include an alpha channel) 1: Use full alpha channel blending mode	W
b5, b4	RDFMT2[1:0]	Texture Format Select Upper	Upper 2 bits of the texture buffer format select bits. See the RDFMT[1:0] bits below for description.	W
b6	BSFA	Alpha Channel Source Blend Factor Select	Valid when the USEACB bit is 1 (full alpha channel blending mode). 0: Use 1.0 as source blend factor for alpha channel 1: Use alpha as source blend factor for alpha channel	W
b7	BDFA	Alpha Channel Destination Blend Factor Select	Valid when the USEACB bit is 1 (full alpha channel blending mode). 0: Use 1.0 as destination blend factor for alpha channel 1: Use alpha as destination blend factor for alpha channel	W
b8	WRFMT2	Frame Buffer Format Select Upper	Upper bit of the frame buffer format select bits. See the WRFMT[1:0] bits below for description.	W
b9	BSF	Source Blend Factor Select	0: Use 1.0 as source blend factor for color channel 1: Use alpha as source blend factor for color channel	W
b10	BDF	Destination Blend Factor Select	0: Use 1.0 as destination blend factor for color channel 1: Use alpha as destination blend factor for color channel	W
b11	BSI	Source Blend Factor Invert	0: Use blend factor as specified through the BSF bit 1: Invert source blend factor (1 – x)	W
b12	BDI	Destination Blend Factor Invert	0: Use blend factor as specified through the BDF bit 1: Invert destination blend factor (1 – x)	W
b13	BC2	Blend Color2 Select	0: Use pixel from frame buffer as destination for color channel 1: Use value of the COLOR2 register as destination for color channel	W
b14	TEXCLPX	U Limiter Texture Clamp Mode Select	This bit specifies the operation of texture mapping in U coordinate (X coordinate in texture space). 0: Mask mode: repeat texture pattern in U direction 1: Clamp mode: repeat last texture data in U direction	W

Bit	Symbol	Bit Name	Description	R/W
b15	TEXCLPY	V Limiter Texture Clamp Mode Select	This bit specifies the operation of texture mapping in V coordinate (Y coordinate in texture space). 0: Mask mode: repeat texture pattern in V direction 1: Clamp mode: repeat last texture data in V direction	W
b16	TEXFILTX	U Axis Texture Filter Enable	0: No filtering on texture U axis 1: Linear filtering on texture U axis	W
b17	TEXFILTY	V Axis Texture Filter Enable	0: No filtering on texture V axis 1: Linear filtering on texture V axis	W
b19, b18	RDFMT[1:0]	Texture Format Select	Pixel format of the texture buffer. b5 b4 b19 b18 0 0 0 0: A(8) (8 bpp) 0 0 0 1: RGB(565) (16 bpp) 0 0 1 0: ARGB(8888) (32 bpp) 0 0 1 1: ARGB(4444) (16 bpp) 0 1 0 0: ARGB(1555) (16 bpp) 0 1 0 1: ACLUT(44), 4-bit alpha and 4-bit indexed color (8 bpp) 1 0 0 1: CLUT(8), 8-bit indexed color (8 bpp) 1 0 1 0: CLUT(4), 4-bit indexed color (4 bpp) 1 0 1 1: CLUT(2), 2-bit indexed color (2 bpp) 1 1 0 0: CLUT(1), 1-bit indexed color (1 bpp) Settings other than above are prohibited.	W
b21, b20	WRFMT[1:0]	Frame Buffer Format Select	Pixel format of the frame buffer. b8 b21 b20 0 0 0: A(8) (8 bpp) 0 0 1: RGB(565) (16 bpp) 0 1 0: ARGB(8888) (32 bpp) 0 1 1: ARGB(4444) (16 bpp) Settings other than above are prohibited.	W
b23, b22	WRALPHA[1:0]	Alpha Source Select	When the USEACB bit is 0 (write-alpha mode): Sets the alpha source for the frame buffer. b23 b22 0 0: Use alpha from the COLOR2 register 0 1: Use source alpha (pixel coverage) 1 0: Use 0.0 as alpha 1 1: Use alpha from frame buffer When the USEACB bit is 1 (full alpha channel blending mode): b23 b22 0 0: Use alpha from the COLOR2 register as destination 0 1: Use alpha from frame buffer as destination 1 0: Use alpha from frame buffer as destination 1 1: Use alpha from frame buffer as destination	W
b24	RLEEN	RLE Enable	0: RLE unit is disabled 1: RLE unit is enabled	W
b25	CLUTEN	CLUT Enable	0: CLUT is disabled, the index is directly put on the internal R, G, B channels 1: CLUT is enabled	W
b26	COLKEYEN	Chroma Keying Enable	0: Chroma keying unit is disabled 1: Chroma keying unit is enabled	W
b27	CLUTFORM	CLUT Format Select	0: ARGB(8888) 1: RGB(565) × 2 (specifies two entries per 32 bits)	W
b28	BSIA	Alpha Channel Source Blend Factor Invert	Valid when the USEACB bit is 1 (full alpha channel blending mode). 0: Use blend factor as specified through the BSFA bit 1: Invert source blend factor (1 – x)	W
b29	BDIA	Alpha Channel Destination Blend Factor Invert	Valid when the USEACB bit is 1 (full alpha channel blending mode). 0: Use blend factor as specified through the BDFA bit 1: Invert destination blend factor (1 – x)	W

Bit	Symbol	Bit Name	Description	R/W
b31, b30	RLEPIXW[1:0]	RLE Texel Width Setting	Specifies the width of a texel for the RLE unit. b31 b30 0 0: 1 byte per texel 0 1: 2 bytes per texel 1 0: 3 bytes per texel 1 1: 4 bytes per texel	W

Note: This register is write-only and reading returns the value of the HWVER register. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.3 Interrupt Control Register (IRQCTL)

Address(es): DRW2D.IRQCTL 000E 30C0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	DLIRCLR	ENUIRCLR	DLIREN	ENUIREN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ENUIREN	Enumeration Finished Interrupt Enable	0: Enumeration finished interrupt is disabled 1: Enumeration finished interrupt is enabled	W
b1	DLIREN	Display List Finished Interrupt Enable	0: Display list finished interrupt is disabled 1: Display list finished interrupt is enabled	W
b2	ENUIRCLR	Enumeration Finished Interrupt Clear	0: Do not clear enumeration finished interrupt 1: Clear enumeration finished interrupt	W
b3	DLIRCLR	Display List Finished Interrupt Clear	0: Do not clear display list finished interrupt 1: Clear display list finished interrupt	W
b31 to b4	—	Reserved	The write value should be 0.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.4 Cache Control Register (CACHECTL)

Address(es): DRW2D.CACHECTL 000E 30C4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	CFLUT X	CENTX	CFLUF X	CENFX
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CENFX	Frame Buffer Cache Enable	0: Disable the frame buffer cache 1: Enable the frame buffer cache	W
b1	CFLUFX	Frame Buffer Cache Flush	0: Do not flush the frame buffer cache 1: Flush the frame buffer cache	W
b2	CENTX	Texture Cache Enable	0: Disable the texture cache 1: Enable the texture cache	W
b3	CFLUTX	Texture Cache Flush	0: Do not flush the texture cache 1: Flush the texture cache	W
b31 to b4	—	Reserved	The write value should be 0.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.5 Status Register (STATUS)

Address(es): DRW2D.STATUS 000E 3000h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	DLIR	ENUIR	DLSTACT	CACHEDTY	BSYWR	BSYENUM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	BSYENUM	Enumeration Unit Busy Flag	0: Enumeration unit idle 1: Enumeration unit busy, new primitive cannot be started	R
b1	BSYWR	Frame Buffer Writeback Busy Flag	0: Frame Buffer writeback finished 1: Frame Buffer writeback busy, frame buffer type cannot be changed	R
b2	CACHEDTY	Frame Buffer Cache Dirty Flag	0: Frame Buffer cache is not dirty 1: Frame Buffer cache is dirty, and frame should not be flipped	R
b3	DLSTACT	Display List Reader Busy Flag	0: Display list reader is idle 1: Display list reader busy, and no direct write access to registers allowed	R
b4	ENUIR	Enumeration Finished Interrupt Flag	0: Enumeration not finished or interrupt disabled 1: Enumeration finished interrupt triggered	R
b5	DLIR	Display List Finished Interrupt Flag	0: Display list not finished or interrupt disabled 1: Display list finished interrupt triggered	R
b31 to b6	—	Reserved	These bits are read as 0.	R

52.2.6 Hardware Version Register (HWVER)

Address(es): DRW2D.HWVER 000E 3004h

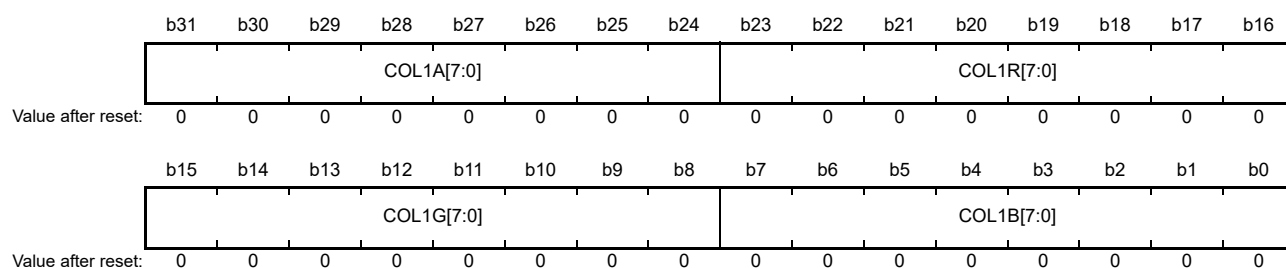
	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	ACBLD	—	COLKEY	TEXCLUT256	RLEUNIT	—	TEXCLUT	PERFCNT	TXCACHE	FBCACHE	DLR	—
Value after reset:	0	0	0	0	1	1	1	1	1	0	1	1	1	1	1	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	REV[11:0]											
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	REV[11:0]	Revision Number	Revision number of DRW2D is stored.	R
b16 to b12	—	Reserved	These bits are read as 0.	R
b17	DLR	Display List Reader Available	0: No display list reader 1: Display list reader is available	R
b18	FBCACHE	Frame Buffer Cache Available	0: No frame buffer cache 1: Frame buffer cache is available	R
b19	TXCACHE	Texture Cache Available	0: No texture cache 1: Texture cache is available	R
b20	PERFCNT	Performance Counter Available	0: No performance counter 1: Two performance counters are available	R
b21	TEXCLUT	Texture CLUT Available	0: No texture CLUT 1: Texture CLUT is available	R
b22	—	Reserved	This bit is read as 0.	R
b23	RLEUNIT	RLE Unit Available	0: No RLE unit 1: RLE unit is available	R
b24	TEXCLUT256	Texture CLUT Size	0: Texture CLUT size is 16 entries 1: Texture CLUT size is 256 entries	R
b25	COLKEY	Chroma Key Available	0: No chroma keying 1: Chroma keying is available	R
b26	—	Reserved	This bit is read as 1.	R
b27	ACBLD	Alpha Channel Blending Available	0: Full alpha channel blending is not available 1: Full alpha channel blending is available	R
b31 to b28	—	Reserved	These bits are read as 0.	R

52.2.7 Base Color Register (COLOR1)

Address(es): DRW2D.COLOR1 000E 3064h

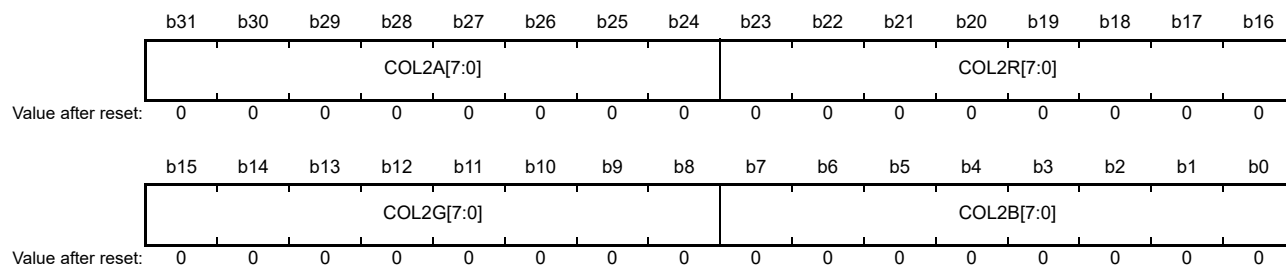


Bit	Symbol	Bit Name	Description	R/W
b7 to b0	COL1B[7:0]	Color1 Blue Channel Setting	Specifies blue channel of color1.	W
b15 to b8	COL1G[7:0]	Color1 Green Channel Setting	Specifies green channel of color1.	W
b23 to b16	COL1R[7:0]	Color1 Red Channel Setting	Specifies red channel of color1.	W
b31 to b24	COL1A[7:0]	Color1 Alpha Channel Setting	Specifies alpha channel of color1. 00h: Transparent ... FFh: Opaque	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.8 Secondary Color Register (COLOR2)

Address(es): DRW2D.COLOR2 000E 3068h

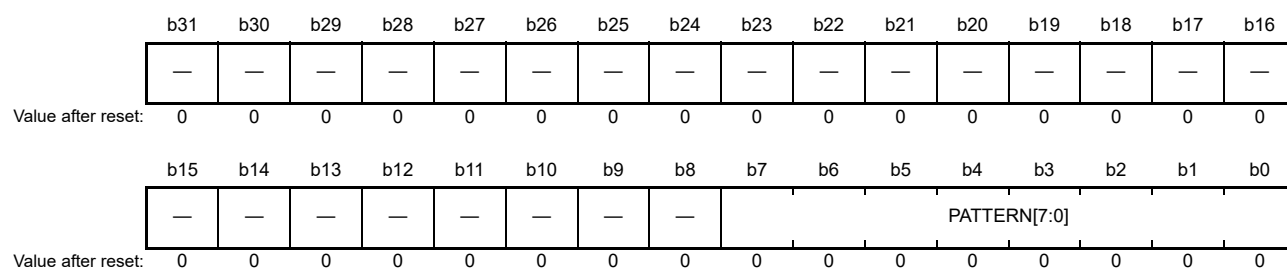


Bit	Symbol	Bit Name	Description	R/W
b7 to b0	COL2B[7:0]	Color2 Blue Channel Setting	Specifies blue channel of color2.	W
b15 to b8	COL2G[7:0]	Color2 Green Channel Setting	Specifies green channel of color2.	W
b23 to b16	COL2R[7:0]	Color2 Red Channel Setting	Specifies red channel of color2.	W
b31 to b24	COL2A[7:0]	Color2 Alpha Channel Setting	Specifies alpha channel of color2. 00h: Transparent ... FFh: Opaque	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.9 Pattern Register (PATTERN)

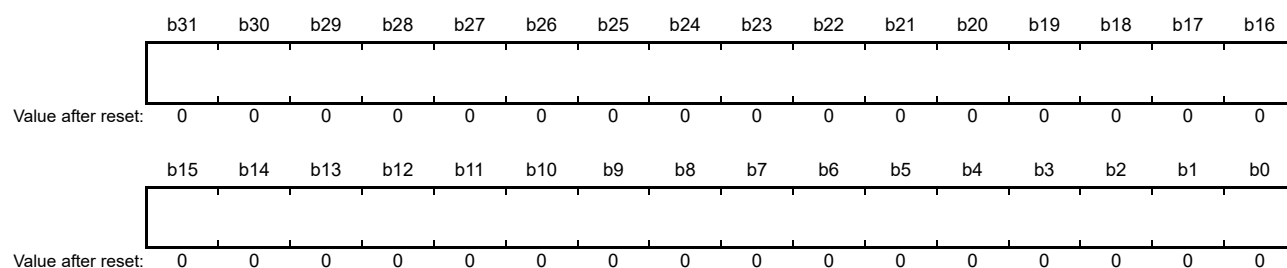
Address(es): DRW2D.PATTERN 000E 3074h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	PATTERN[7:0]	Pattern Bitmap Setting	Specifies bitmap of the pattern. Each bit is interpreted as a reference to one of the two color registers (0: COLOR1, 1: COLOR2)	W
b31 to b8	—	Reserved	The write value should be 0.	W

52.2.10 Limiter n Start Value Register (LnSTART) (n = 1 to 6)

Address(es): DRW2D.L1START 000E 3010h, DRW2D.L2START 000E 3014h, DRW2D.L3START 000E 3018h, DRW2D.L4START 000E 301Ch, DRW2D.L5START 000E 3020h, DRW2D.L6START 000E 3024h



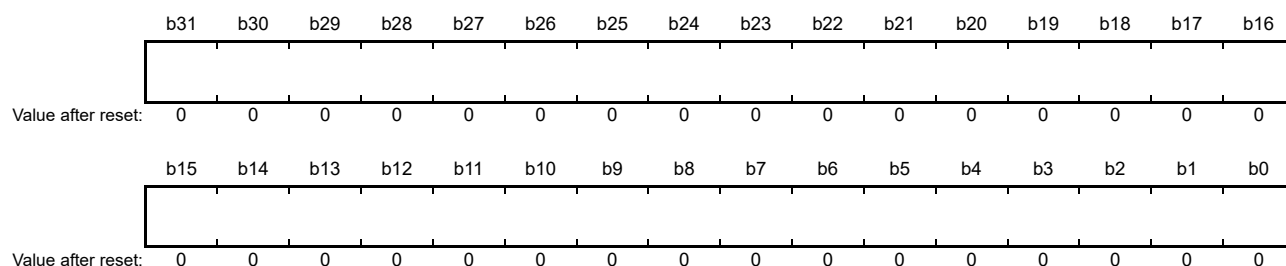
This register specifies the start value of limiter n (n = 1 to 6) used for rasterization.

The start value is a 16:16 fixed point number valid at the first pixel of the bounding box.

The LnSTART registers are write-only and read as undefined.

52.2.11 Limiter n X-Axis Increment Register (LnXADD) (n = 1 to 6)

Address(es): DRW2D.L1XADD 000E 3028h, DRW2D.L2XADD 000E 302Ch, DRW2D.L3XADD 000E 3030h,
DRW2D.L4XADD 000E 3034h, DRW2D.L5XADD 000E 3038h, DRW2D.L6XADD 000E 303Ch



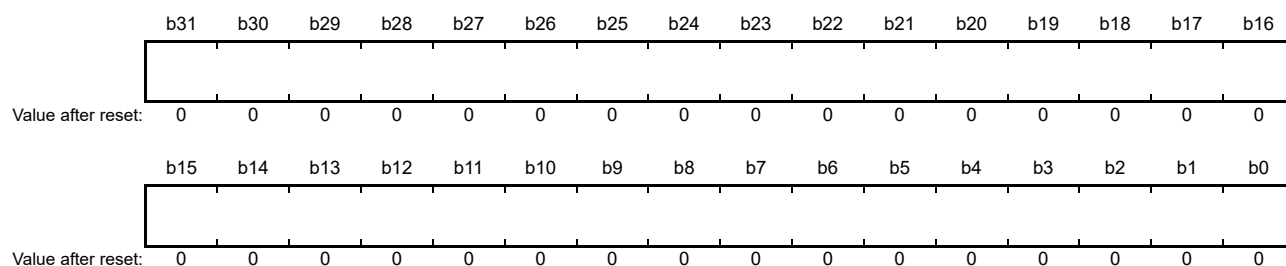
This register specifies the X-axis increment value of limiter n (n = 1 to 6) used for rasterization.

The X-axis increment value is the 16:16 fixed point difference between two samples with a distance of 1 pixel along the X axis.

The LnXADD registers are write-only and read as undefined.

52.2.12 Limiter n Y-Axis Increment Register (LnYADD) (n = 1 to 6)

Address(es): DRW2D.L1YADD 000E 3040h, DRW2D.L2YADD 000E 3044h, DRW2D.L3YADD 000E 3048h,
DRW2D.L4YADD 000E 304Ch, DRW2D.L5YADD 000E 3050h, DRW2D.L6YADD 000E 3054h



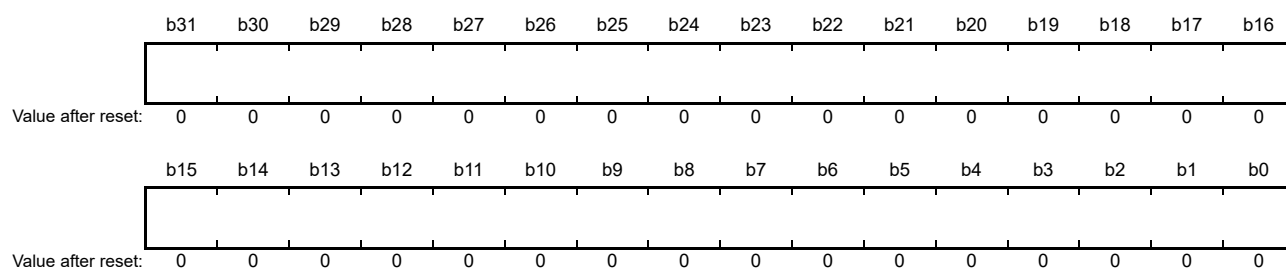
This register specifies the Y-axis increment value of limiter n (n = 1 to 6) used for rasterization.

The Y-axis increment value is the 16:16 fixed point difference between two samples with a distance of 1 pixel along the Y axis.

The LnYADD registers are write-only and read as undefined.

52.2.13 Limiter m Band Width Parameter Register (LmBAND) (m = 1, 2)

Address(es): DRW2D.L1BAND 000E 3058h, DRW2D.L2BAND 000E 305Ch



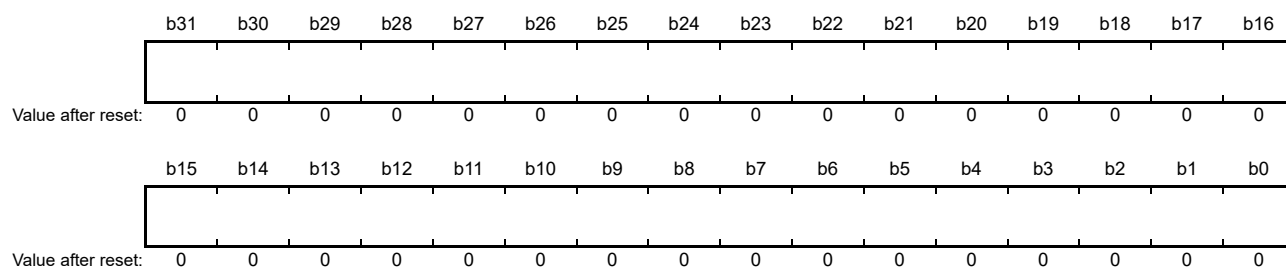
This register specifies the inner width of band region when the CONTROL.BANDmEN bit (m = 1, 2) is 1 (band output filtering for limiter m is enabled).

The band width is the 16:16 fixed point number.

The LmBAND registers are write-only and read as undefined.

52.2.14 Texture Base Address Register (TEXORG)

Address(es): DRW2D.TEXORG 000E 30BCh

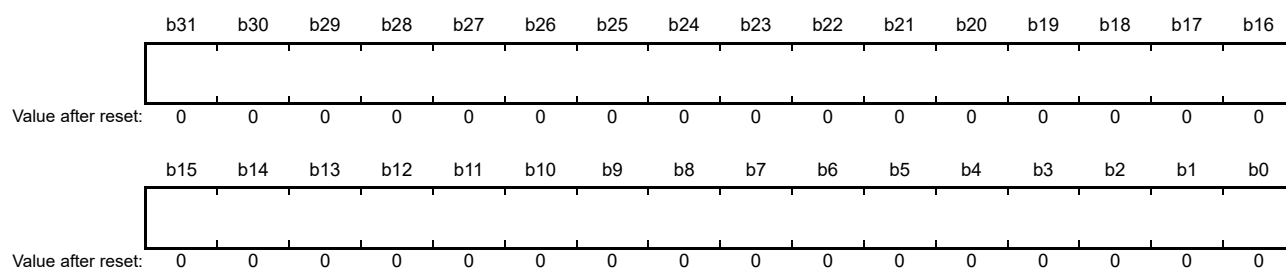


This register specifies the address of the upper left corner texel (texture base address).

The TEXORG register is write-only and read as undefined.

52.2.15 Texels Per Texture Line Register (TEXPITCH)

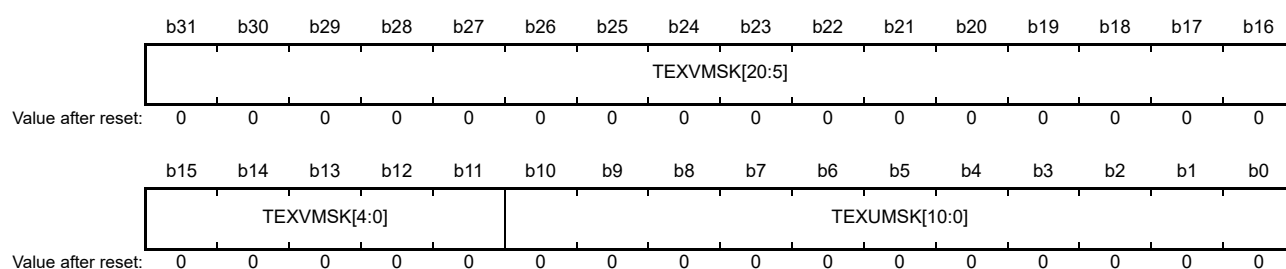
Address(es): DRW2D.TEXPITCH 000E 30B4h



This register specifies the texels per texture line. The valid range is 0 to 2047.
The TEXPITCH register is write-only and read as undefined.

52.2.16 Texture Mask Register (TEXMSK)

Address(es): DRW2D.TEXMSK 000E 30B8h



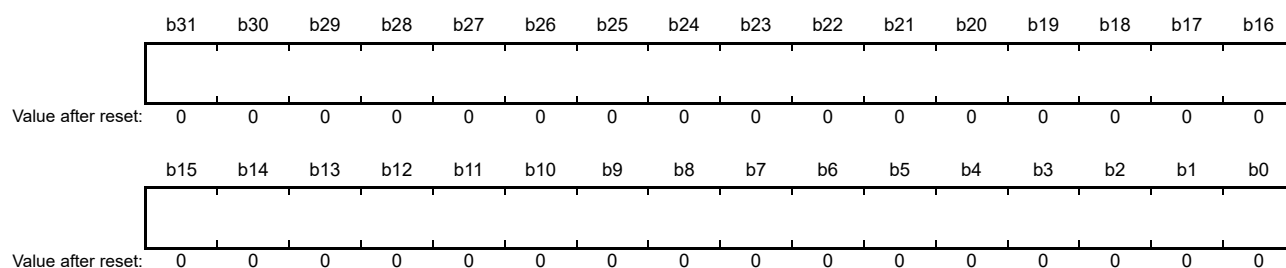
Bit	Symbol	Bit Name	Description	R/W
b10 to b0	TEXUMSK[10:0]	U Mask Setting	When the CONTROL2.TEXCLPX bit is 0 (mask mode): These bits function as mask bits in U coordinate. Set to power of 2 – 1. When the CONTROL2.TEXCLPX bit is 1 (clamp mode): Set to texture width – 1. The maximum value of texture width is 2048.	W
b31 to b11	TEXVMSK[20:0]	V Mask Setting	When the CONTROL2.TEXCLPX bit is 0 (mask mode): These bits function as mask bits in V coordinate. Set to power of 2 – 1. When the CONTROL2.TEXCLPX bit is 1 (clamp mode): Set to texture height – 1. The maximum value of texture height is 1024.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

This register is used to specify the texture size when the CONTROL2.TEXCLPX bit is 1 or specify masks for U and V separately to wraparound values of U and V when the TEXCLPX bit is 0.

52.2.17 U Limiter Start Value Register (LUST)

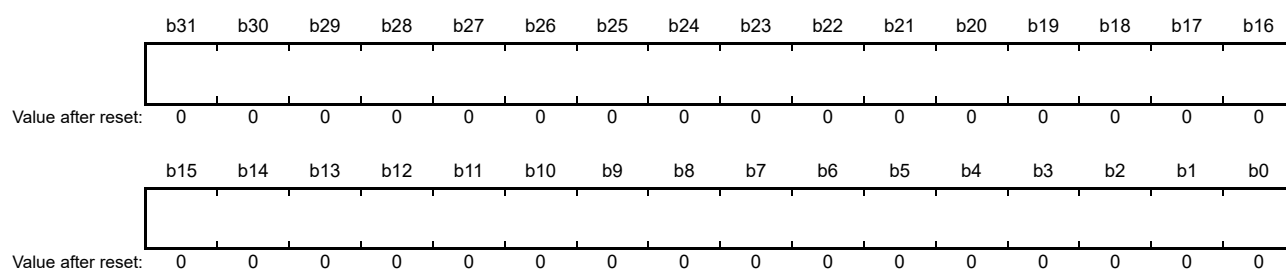
Address(es): DRW2D.LUST 000E 3090h



This register specifies the start value of U limiter used for texture mapping.
 The start value is a 16:16 fixed point number valid at the first pixel of the bounding box.
 The LUST register is write-only and read as undefined.

52.2.18 U Limiter X-Axis Increment Register (LUXADD)

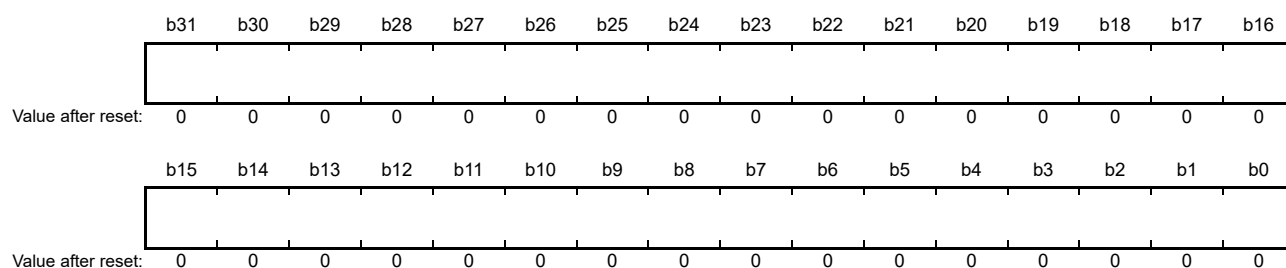
Address(es): DRW2D.LUXADD 000E 3094h



This register specifies the X-axis increment value of U limiter used for texture mapping.
 The X-axis inclement value is the 16:16 fixed point difference between two samples with a distance of 1 pixel along the X axis.
 The LUXADD register is write-only and read as undefined.

52.2.19 U Limiter Y-Axis Increment Register (LUYADD)

Address(es): DRW2D.LUYADD 000E 3098h



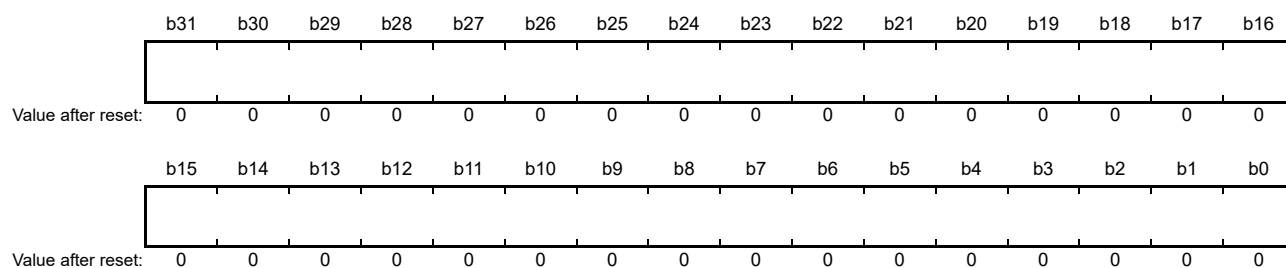
This register specifies the Y-axis increment value of U limiter used for texture mapping.

The Y-axis increment value is the 16:16 fixed point difference between two samples with a distance of 1 pixel along the Y axis.

The LUYADD register is write-only and read as undefined.

52.2.20 V Limiter Start Value Integer Part Register (LVSTI)

Address(es): DRW2D.LVSTI 000E 309Ch



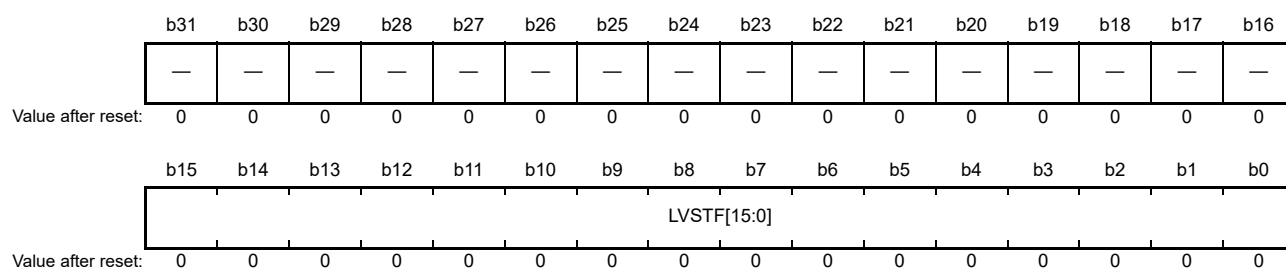
This register specifies the integer part of the start value of V limiter used for texture mapping.

The start value is a 32:16 fixed point number valid at the first pixel of the bounding box.

The LVSTI register is write-only and read as undefined.

52.2.21 V Limiter Start Value Fractional Part Register (LVSTF)

Address(es): DRW2D.LVSTF 000E 30A0h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	LVSTF[15:0]	V Limiter Start Value Fractional Part Setting	Specifies fractional part of V limiter start value.	W
b31 to b16	—	Reserved	The write value should be 0.	W

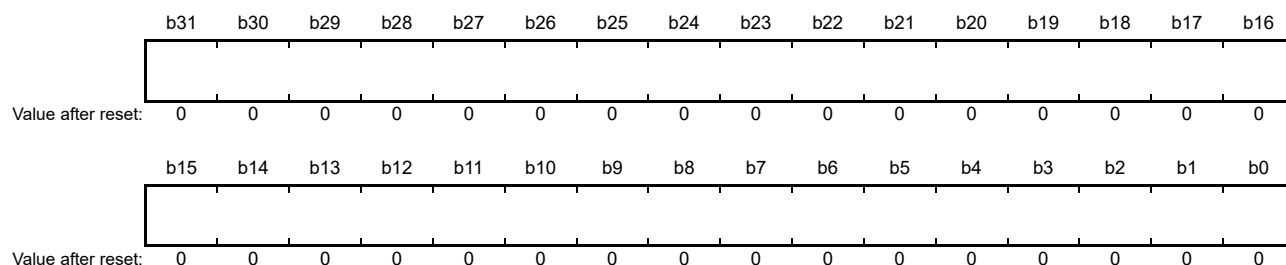
This register specifies the fractional part of the start value of V limiter used for texture mapping.

The start value is a 32:16 fixed point number valid at the first pixel of the bounding box.

The LVSTF register is write-only and read as undefined.

52.2.22 V Limiter X-Axis Increment Integer Part Register (LVXADDI)

Address(es): DRW2D.LVXADDI 000E 30A4h



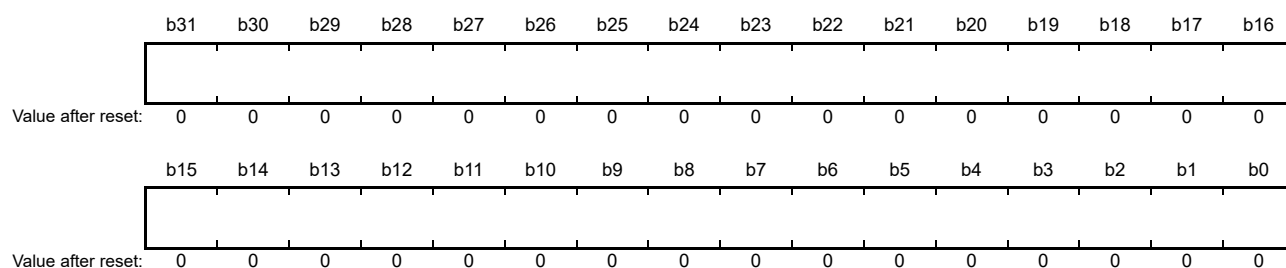
This register specifies the integer part of X-axis increment value of V limiter used for texture mapping.

The X-axis increment value is the 32:16 fixed point difference between two samples with a distance of 1 pixel along the X axis.

The LVXADDI register is write-only and read as undefined.

52.2.23 V Limiter Y-Axis Increment Integer Part Register (LVYADDI)

Address(es): DRW2D.LVYADDI 000E 30A8h



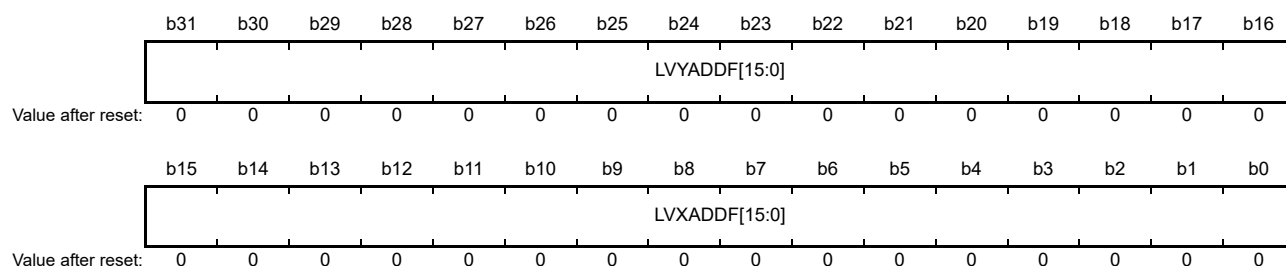
This register specifies the integer part of Y-axis increment value of V limiter used for texture mapping.

The Y-axis inclement value is the 32:16 fixed point difference between two samples with a distance of 1 pixel along the Y axis.

The LVYADDI register is write-only and read as undefined.

52.2.24 V Limiter Increment Fractional Parts Register (LVYXADDF)

Address(es): DRW2D.LVYXADDF 000E 30ACh



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	LVXADDF[15:0]	V Limiter X-Axis Increment Fractional Part Setting	Specifies fractional part of V limiter X-axis increment.	W
b31 to b16	LVYADDF[15:0]	V Limiter Y-Axis Increment Fractional Part Setting	Specifies fractional part of V limiter Y-axis increment.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

This register specifies both fractional parts of X-axis and Y-axis increment values of V limiter used for texture mapping.

52.2.25 CLUT Start Address Register (TEXCLADDR)

Address(es): DRW2D.TEXCLADDR 000E 30DCh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0	
	—	—	—	—	—	—	—	—	CLADDR[7:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CLADDR[7:0]	CLUT Start Address Setting	Specifies the write starting address of CLUT for the indexed texture format	W
b31 to b8	—	Reserved	The write value should be 0.	W

52.2.26 CLUT Data Register (TEXCLDATA)

Address(es): DRW2D.TEXCLDATA 000E 30E0h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

This register is used to set the write data to the address specified by the TEXCLADDR register.

When the CONTROL2.CLUTFORM bit is 0 (ARGB(8888)), write CLUT data to b31 to b0. When 1 (RGB(565)), write data in 32-bit unit by setting CLUT data for even index to b15 to b0 and CLUT data for odd index to b31 to b16.

After writing each data, the TEXCLADDR register is incremented by 1.

The TEXCLDATA register is write-only and read as undefined.

52.2.27 CLUT Offset Register (TEXCLOFST)

Address(es): DRW2D.TEXCLOFST 000E 30E4h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0	
	—	—	—	—	—	—	—	—	CLOFST[7:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Bit	Symbol	Bit Name	Description	R/W
b7 to b0	CLOFST[7:0]	CLUT Offset Setting	Specifies CLUT offset. The CLOFST[7:0] bits are combined with the original index by a bitwise OR operation.	W
b31 to b8	—	Reserved	The write value should be 0.	W

This register is used to specify the offset to the texture index for the indexed texture formats CLUT(8), CLUT(4), CLUT(2), CLUT(1), and ACLUT(44) to allow selecting an offset part of the CLUT.

52.2.28 Chroma Key Register (COLKEY)

Address(es): DRW2D.COLKEY 000E 30E8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16	
	—	—	—	—	—	—	—	—	R[7:0]								
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	G[7:0]								B[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

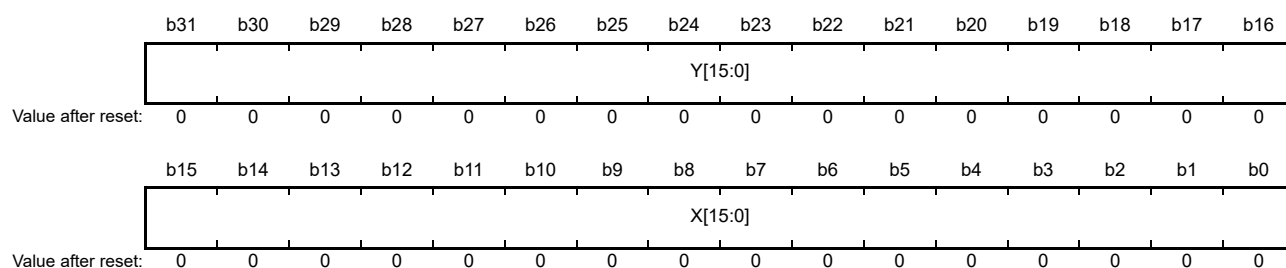
Bit	Symbol	Bit Name	Description	R/W
b7 to b0	B[7:0]	Chroma Key Blue Channel Setting	Specifies blue channel of chroma key.	W
b15 to b8	G[7:0]	Chroma Key Green Channel Setting	Specifies green channel of chroma key.	W
b23 to b16	R[7:0]	Chroma Key Red Channel Setting	Specifies red channel of chroma key.	W
b31 to b24	—	Reserved	The write value should be 0.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

The R, G, and B components of the internal color representation of a texel are compared with the chroma key. If the values are equal then A, R, G, and B are set to 00h (transparent), else A is set to FFh.

52.2.29 Bounding Box Dimension Register (SIZE)

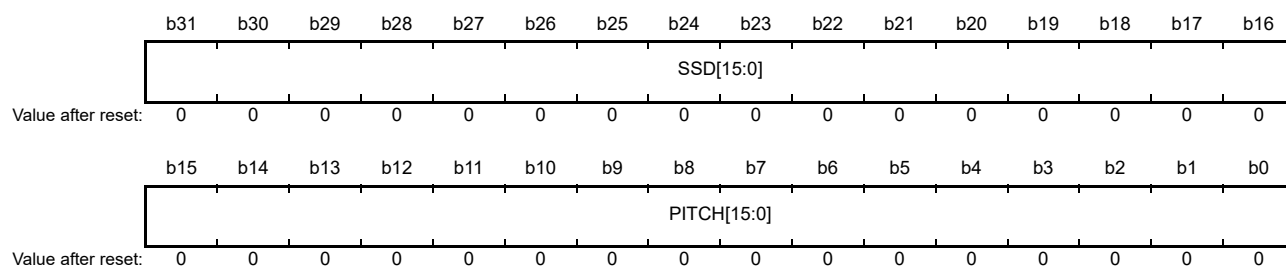
Address(es): DRW2D.SIZE 000E 3078h



Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.30 Frame Buffer Pitch Register (PITCH)

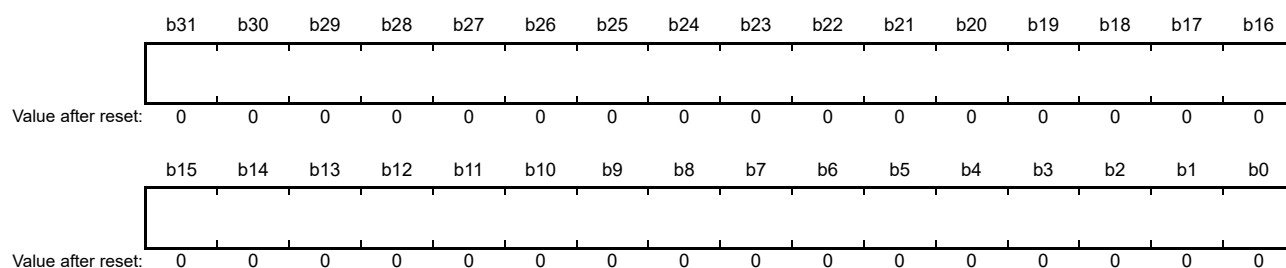
Address(es): DRW2D.PITCH 000E 307Ch



Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.31 Frame Buffer Base Address Register (ORIGIN)

Address(es): DRW2D.ORIGIN 000E 3080h



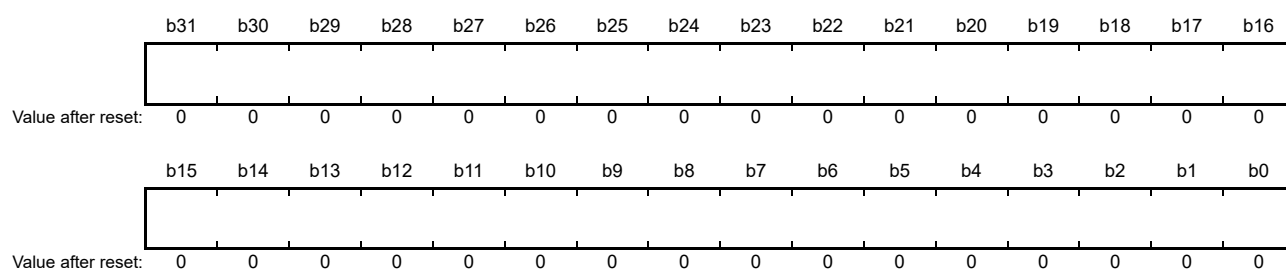
This register specifies the address of the first pixel in frame buffer (frame buffer base address).

Writing to this register triggers DRW2D to start rendering.

The ORIGIN register is write-only and read as undefined.

52.2.32 Display List Start Address Register (DLISTST)

Address(es): DRW2D.DLISTST 000E 30C8h



This register specifies the start address of display list.

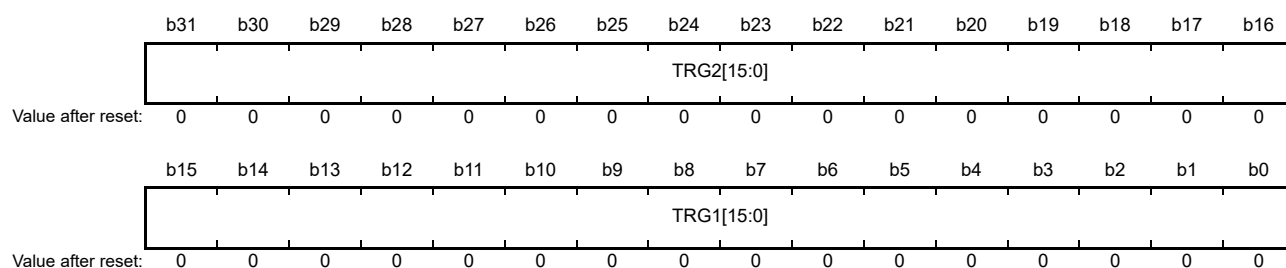
Setting a new display list start address to this register triggers execution of the new display list.

When setting a new display list start address to this register, confirm that the execution of the previous display list is finished.

The DLISTST register is write-only and read as undefined.

52.2.33 Performance Counters Control Register (PERFTRG)

Address(es): DRW2D.PERFTRG 000E 30D4h

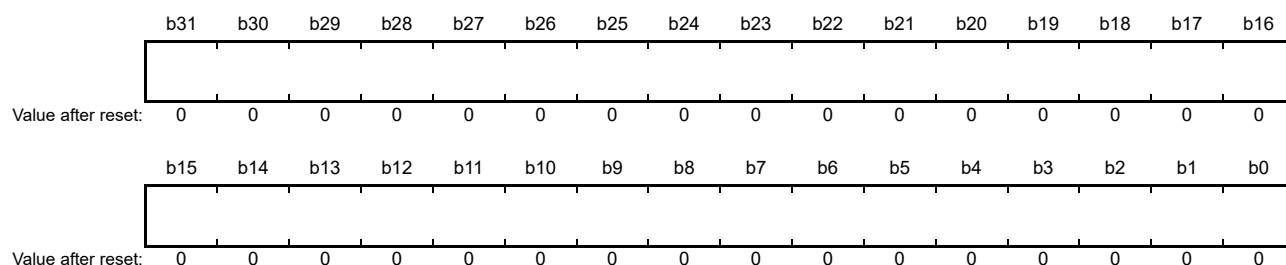


Bit	Symbol	Bit Name	Description	R/W
b15 to b0	TRG1[15:0]	Performance Counter 1 Trigger Setting	Selects the internal event that increments the PERFCNT1 register. 0: Performance counter 1 is disabled 1: DRW2D active cycles 2: Frame buffer read access 3: Frame buffer write access 4: Texture read access 5: Invisible pixels (enumerated but selected with alpha 0%) 6: Invisible pixels while internal FIFO is empty (lost cycles) 7: Display list reader active cycles 8: Frame buffer read hits 9: Frame buffer read misses 10: Frame buffer write hits 11: Frame buffer write misses 12: Texture read hits 13: Texture read misses 31: Every clock cycle (for use as timer) Settings other than above are prohibited.	W
b31 to b16	TRG2[15:0]	Performance Counter 2 Trigger Setting	Same as for the TRG1[15:0] bits, but for the PERFCNT2 register.	W

Note: This register is write-only and read as undefined. Do not use a bit processing instruction but the MOV instruction to set this register.

52.2.34 Performance Counter k (PERFCNTk) (k = 1, 2)

Address(es): DRW2D.PERFCNT1 000E 30CCh, DRW2D.PERFCNT2 000E 30D0h



Reading this register returns the actual value of the performance counter that is incremented on every event selected by the PERFTRG register.

Writing “0000 0000h” to this register clears the performance counter.

52.3 Drawing Features

52.3.1 Drawing Features Summary

52.3.1.1 Color Formats

Supported color formats are:

Frame Buffer formats

- 8-bit: A(8)
- 16-bit: RGB(565), ARGB(4444)
- 32-bit: ARGB(8888)

Texture formats

- 1-bit: CLUT(1)
- 2-bit: CLUT(2)
- 4-bit: CLUT(4)
- 8-bit: A(8), CLUT(8), ACLUT(44)
- 16-bit: ARGB(4444), ARGB(1555), RGB(565)
- 24-bit: RGB(888) (RLE unit)
- 32-bit: ARGB(8888)

CLUT formats use a 256-entry color lookup table.

52.3.1.2 Vector Drawing Features

The vector drawing engine uses a half-plane rendering approach, which simplifies implementation of edge anti-aliasing and blurring features without much overhead. When combining some of its functional units, the module can draw not only linear primitives such as lines or polygons, but also quadratic equation-based primitives such as circles and ellipsoids. The following primitives and texture mapping are supported:

- Lines
- Polygons
- Circles and ellipses
- Quadratic curves (software driver support)
- 2D texture mapping
- Bilinear filtering of the textures

52.3.1.3 BitBLT Features

The DRW2D supports the BitBLT features using its vector drawing function to draw a rectangle and texture it based on the selected BitBLT function. This approach results in the following BitBLT features:

- Fill
- Copy
- Stretch BitBLT
- Rotate and scale
- Alpha blending
- Bilinear filtering
- Color conversion

52.3.2 Vector Drawing

Supported vector drawing includes:

Lines

- Arbitrary width
- Round endpoints
- Truncated endpoints
- Alpha gradients
- Soft edges (blurring)
- Render attribute: color, pattern, or texture

Polygons

- Triangles and quadrangles (complex polygons are tessellated by software)
- Alpha gradients
- Soft edges (blurring)
- Per edge controls for anti-aliasing
- Render attribute: color, pattern, or texture

Circles and ellipses

- All conic sections
- Filled or with arbitrary width
- Arcs of 0° to 360°
- Soft edges
- Alpha gradients
- Render attribute: color, pattern, or texture

Quadratic Béziers

- Approximated by circle arcs
- Arbitrary width
- Round or truncated endpoints
- Outlines, blurring
- Alpha gradients
- Render attribute: color

Texture mapping

- 2D array of pixels that can be mapped implicitly or explicitly on all primitives provided by the DRW2D
- Translation, rotation, and scaling/shearing
- Bilinear filtering of the textures
- 3D-like texturing accomplished with line-by-line mapping, if constant in one axis

52.4 Rendering Modes

The rendering process can be performed in two different modes, register mode and display list mode.

52.4.1 Register Mode

In register mode, when operation is based on register settings, the CPU configures and initiates each render process separately. To start a new render process, the CPU must wait until the previous one is completed. In this mode, the CPU is heavily engaged throughout the entire drawing procedure and is consequently to a large extent unavailable for other tasks.

The CPU must set up all registers for performing a certain drawing operation before it can start the rendering process. A new register setup can only be started when the previous render process has completed. Before starting a new register setup, make sure that:

- STATUS.DLSTACT = 0: Display list reader is idle
- STATUS.BSYENUM = 0: Pixel selection unit is idle.

Lastly, write the frame buffer start address to the ORIGIN register. This write triggers the DRW2D to start rendering.

52.4.2 Display List Mode

In display list mode, the CPU creates a display list in memory prior to starting the DRW2D. Such a display contains a bundle of render operations. When started, the DRW2D executes the display list autonomously in parallel with the CPU, which is not involved with drawing operations most of the time. Use of a display list allows a fully asynchronous operation of the CPU and the DRW2D and offers the best possible system performance.

In this mode, the display list reader reads a memory block containing instructions on how to set the DRW2D control registers and executes these control register writes accordingly.

52.5 Interrupts

The DRW2D generates two interrupts:

- DRWENUIR interrupt
- DRWDLIR interrupt

52.5.1 Interrupt Sources

(1) DRWENUIR Interrupt

This is the current render process finished interrupt.

(2) DRWDLIR Interrupt

This is the display list finished interrupt. It is asserted on completion of a display list process. DRWDLIR is activated if one of the following is true:

- The entire display list is complete
- The display list processing stops because a new display list start is triggered by a write to the DLISTST register.

52.5.2 Interrupt Control

The two DRW2D interrupts are combined into a single shared interrupt, DRWIRQ, to the CPU. Each interrupt can be disabled or enabled by setting its associated enable bit in the IRQCTL register.

The occurrence of an enabled interrupt (one with its enable bit set to 1 in the IRQCTL register) is monitored in the STATUS register with its associated interrupt status bit is set to 1. The shared DRW2D interrupt DRWIRQ is then generated.

To clear the interrupt, write 1 to the interrupt clear bit in the IRQCTL register. The interrupt clear bit returns to 0 automatically.

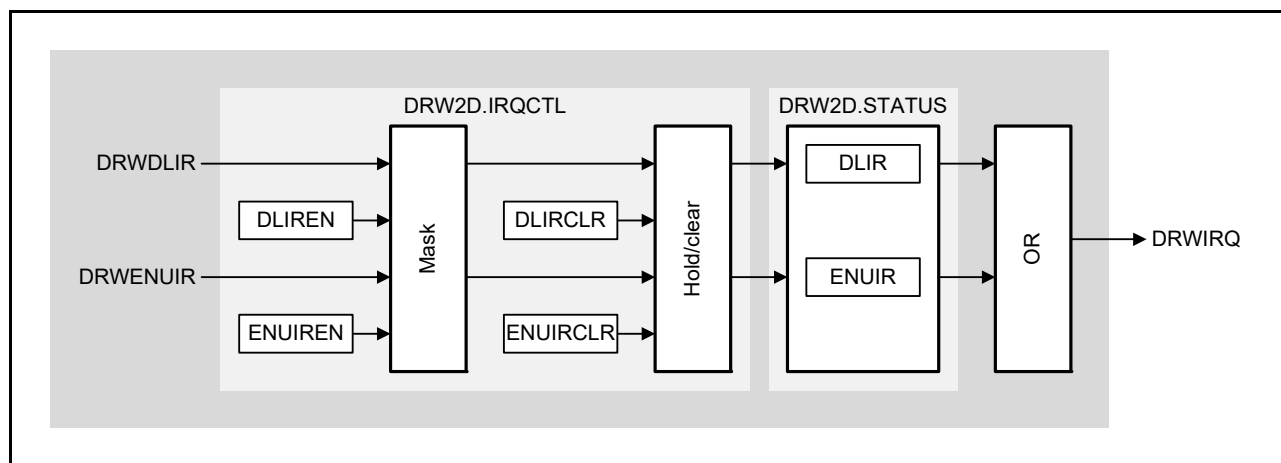


Figure 52.3 Interrupt Controller Unit

52.6 Usage Notes

52.6.1 Notes for Endian Setting

DRW2D accesses a memory on the premise that all data are stored in little-endian format. When using the DRW2D, set the MDE.MDE[2:0] bits to 111b (little endian). Refer to [section 7, Option-Setting Memory \(OFSM\)](#) for details on the MDE register.

52.6.2 DRW2D Library

Use of the DRW2D requires the DRW2D library provided by Renesas Electronics. Please contact our sales office for information regarding the DRW2D library.